

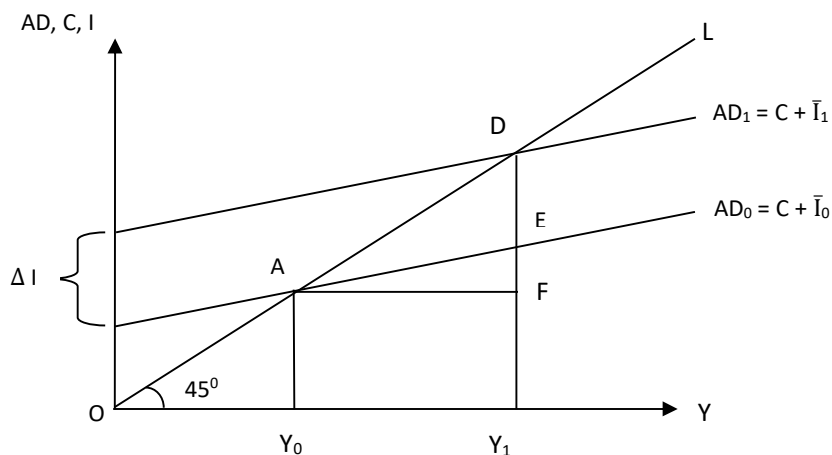


Fig. 2.5: Investment Multiplier

The initial equilibrium in Fig. 2.5 occurs at A where $AD_0 (= C + \bar{I}_0)$ intersects OL. With an increase in autonomous investment to $\bar{I}_1$ (i.e., $\Delta I = \bar{I}_1 - \bar{I}_0$), the AD curve shifts upward to $AD_1 (= C + \bar{I}_1)$ and the new equilibrium occurs at D where AD_1 intersects OL. Note that while investment increased by $\Delta I (= DE)$, the change in income $\Delta Y (= DF)$ that it brings about is greater, i.e., $\Delta Y > \Delta I$ (where, $\Delta Y = Y_1 - Y_0 = AF = DF$ and $DF > DE = \Delta I$), indicating the operation of a multiplier effect.

Induced Investment

Suppose the investment function is given by equation (2.2b) above, i.e., $I = c + dY$. In this case, investment demand has an induced component that is affected by changes in income. The equilibrium income would be derived as follows:

$Y = AD = a + bY + c + dY$, so that:

$$Y_e = (a + c) / [1 - (b + d)] \quad \dots (2.7)$$

ceteris paribus, if there is a change in autonomous investment in this case (i.e., $\Delta I = \Delta c$; $\Delta a = 0$), the multiplier would be given by: $1 / [1 - (b + d)]$. This would follow since $\Delta Y = \{1 / [1 - (b + d)]\} \Delta c$. Note that in this case we assumed that $(b + d) < 1$ (i.e., the sum of MPC and MPI is less than one).

2.3.3 Government Expenditure and Tax Multipliers

We now consider the case of a three-sector economy. For simplicity we assume that investment demand is autonomous and the government incurs a constant amount of expenditure G which is autonomous. The government also collects taxes T . We will consider two cases: (i) the government collects a lump sum tax T , and (ii) the government sets a proportional tax rate t , so that its tax revenue $T = tY$. The government's budget deficit (B) is defined as $B = G - T - TR$, where TR refers to transfer payments made by the government. In what follows we will assume, for simplicity that $TR = 0$.

Lump Sum Taxes

When the government imposes a lump sum tax T , the equilibrium income would be given by (2.4) above: $Y_e = (a - bT + \bar{I} + G) / (1-b)$. In this case, the multiplier for any change in autonomous expenditure (either investment or government expenditure) would be given by: $1 / (1 - b)$. Thus, $\Delta Y / \Delta G = 1 / (1 - b)$ or $\Delta Y / \Delta I = 1 / (1 - b)$. If there is an increase in the amount of tax collected, then income would fall. In such a case the tax multiplier would be given by: $-b / (1 - b)$. Thus, $\Delta Y / \Delta T = -b / (1 - b)$. Recall that taxes are a leakage from the income stream, so increase in taxes leads to a decrease in disposable income. Consequently, there is a decrease in consumption demand, leading to a decline in output (as output is demand-determined, lower demand results in lowering of output). You should note that there is a two-way relationship: consumption is influenced by the level of income (evident from the consumption function) and income is influenced by consumption demand.

Balanced Budget Multiplier

Suppose the government increases its spending (i.e., $\Delta G > 0$) and cut taxes by exactly the same amount ($\Delta T < 0$). It implies that the government budget remains unchanged. If ΔB indicates budget balance, then $\Delta B = \Delta G - \Delta T = 0$. Alternatively we can write $\Delta G = \Delta T$. You may think that such action would have no impact on equilibrium income as the injection of additional demand (ΔG) is exactly offset by a leakage from the demand stream ($\Delta G = \Delta T$). However, we can show that the multiplier in this case is equal to one, i.e. income increases by the amount of increase in government expenditure. From equation (2.4) we know that $Y_e = (a - bT + \bar{I} + G) / (1-b)$, so that $\Delta Y = (-b\Delta T + \Delta G) / (1-b)$. Since $\Delta G = \Delta T$, this can be written as $\Delta Y = (1-b)\Delta G / (1-b)$, or $\Delta Y / \Delta G = 1$.

Proportional Taxes

Now suppose the government levies a proportional tax on income at tax rate t , where $0 < t < 1$. Since $AD = a + b(Y - tY) + \bar{I} + G$, the equilibrium condition is given by

$$Y = a + b(1 - t)Y + \bar{I} + G \quad \dots (2.7)$$

By re-arranging terms, we obtain the equilibrium level of output as

$$Y_e = (a + \bar{I} + G) / \{1 - b(1 - t)\} \quad \dots (2.8)$$

Equation (2.8) implies that the autonomous expenditure multiplier (for a change in investment or government expenditure) is now lower compared to the cases without the proportional tax. Without proportional taxes the multiplier is given by $1/(1 - b)$. Now, in the presence of proportional taxes, it is given by $\Delta Y / \Delta G = \Delta Y / \Delta I = 1 / \{1 - b(1 - t)\}$. The reduction in the value of the multiplier arises because, out of the increase in income in each round, a fraction t has to be paid in taxes.

You can see from equation (2.8) that any increase in the tax rate would lower equilibrium income, whereas any decrease in tax rate would have the opposite

effect. Therefore, you should see the logic behind tax cuts offered by governments to revive economic activity. However, remember that our assumption is that there is excess production capacity, which is the main reason why the increase in consumption demand spurred by a cut in taxes would lead to an increase in output. Else, it will lead to price rise instead of increase in output.

2.3.4 Open Economy Multiplier

In an open economy there is an additional source of demand for domestic output, viz., exports (X). Exports are autonomous as they depend on foreign incomes (i.e., incomes of foreign residents who buy domestic products) rather than domestic income. Therefore, an increase in exports will have multiplier effect on income, via various rounds of induced increases in consumption. Firms and households purchase goods and services from foreign countries in an open economy. Such imports (M) are a leakage from the domestic expenditure and income streams. We assume that demand for imports depend on domestic income level. The import demand function is given by $M = h + mY$, where h is the autonomous component of import demand. Here, m is the *marginal propensity to import* (MPM) and $m > 0$. Note that both exports and imports are influenced by exchange rate changes also (we will take up such issues later in the course on International Economics). In the present Unit we consider any change in exports and imports due to exchange rate fluctuation is treated as an autonomous change.

In an open economy, exports are an injection while imports are a leakage from the stream of aggregate demand for domestic goods. Thus, aggregate demand is given by: $AD = C + \bar{I} + G + X - M = a + bY + \bar{I} + G + X - h - mY$. Therefore, the equilibrium condition is given by

$$Y = a + bY + \bar{I} + G + X - h - mY \quad \dots (2.9)$$

The equilibrium level of output or income is given by

$$Y_e = (a - h + \bar{I} + G + X) / (1 - b + m) \quad \dots (2.10)$$

In (2.10), the autonomous components are I , G and X . Any change in these autonomous components will have a multiplier effect. The autonomous expenditure multiplier for investment, government expenditure or exports is given by: $\Delta Y / \Delta G = \Delta Y / \Delta I = \Delta Y / \Delta X = 1 / (1 - b + m)$. Clearly, the open economy multiplier is smaller compared to the one for the closed economy ($1/(1 - b)$). The reason is that a proportion m out of the increment in income is now spent on imports and domestic demand is lower to that extent as compared to a closed economy.

In general, from our discussion on multipliers you should note the following points: (i) the multiplier would be larger, the larger is the MPC (b) and MPI (d), the lower the tax rate (t) and the smaller the m ; (ii) the multiplier is applicable both in case of a rise as well as a fall in autonomous incomes, meaning that income would fall by a multiple in case of a given cut in any autonomous component of demand; and (iii) the operation of the multiplier is driven by the